

TC2006B: Device Communications

Transport Layer I: Process-to-Process Delivery

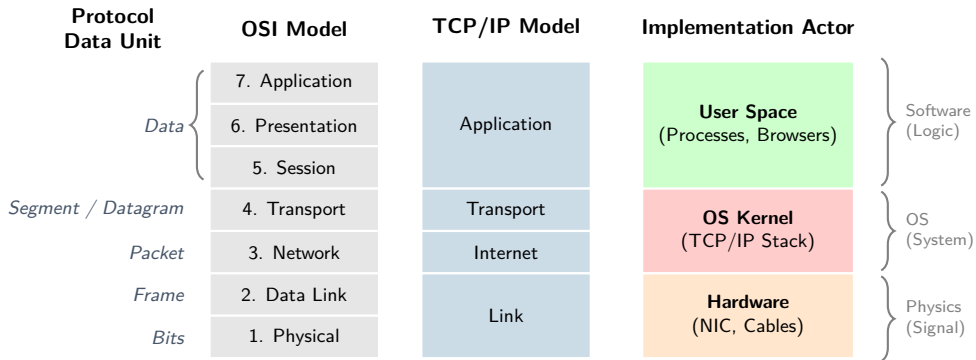
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Today's Agenda

- 1 Transport Layer Basics
- 2 Ports & Protocols
- 3 TCP Mechanisms
- 4 Real-World Applications

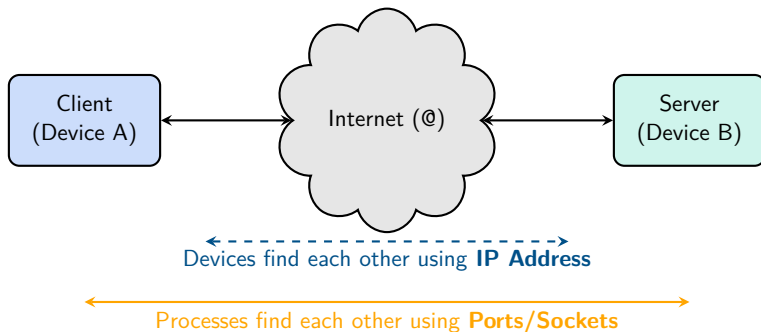
OSI and TCP/IP Models



Device vs. Process Communication

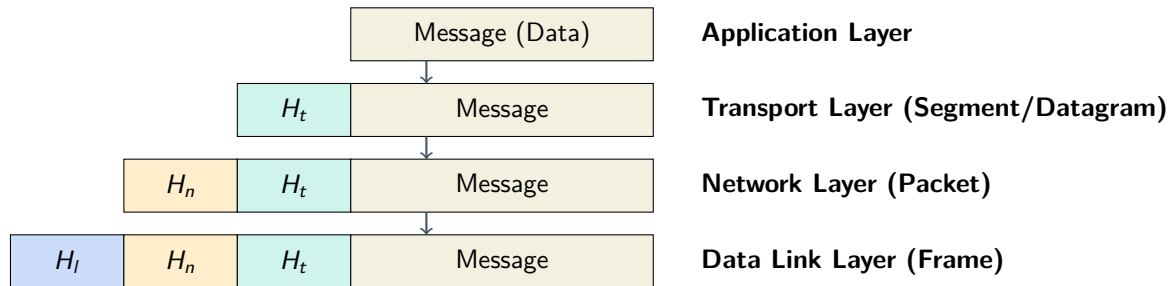
The Goal of the Transport Layer

The transport layer aims to provide **end-to-end delivery** between **processes**, rather than just devices.



Data Encapsulation

As data moves down the stack, headers are appended to the payload.



Addressing Processes

An **endpoint/process** in the network is identified by a socket, also known as a port.

IP Address + Transport Protocol + Port Number

- **OS Administration:** The Operating System manages ports, binding specific processes to them and routing incoming traffic accordingly (e.g., restricting well-known ports to Admin/root).
- **Protocol Separation:** A port is isolated by its transport protocol. This means TCP port 53 and UDP port 53 are completely independent endpoints.

IANA Port Ranges

- **0 to 1023:** Well-known (Privileged)
- **1024 to 49151:** Registered
- **49152 to 65535:** Dynamic / Ephemeral

Common Well-Known Ports

- **FTP:** 21 **SSH:** 22
- **HTTP:** 80 **HTTPS:** 443
- **DNS:** 53 (Uses both TCP & UDP)
- **DHCP:** 67, 68 (Uses UDP)

TCP vs. UDP Overview

The Transport layer is primarily dominated by two protocols, choosing between speed and reliability.

TCP (Transmission Control Protocol)

- **Connection-oriented:** Establishes a session before sending data.
- **Reliable:** Guarantees delivery, order, and error-checking.
- **Slower:** Higher overhead due to ACKs and handshakes.
- *Use case: Web browsing, file transfer, emails.*

UDP (User Datagram Protocol)

- **Connectionless:** Sends data immediately without establishing a session.
- **Unreliable:** "Fire and forget." No guarantee of delivery or order.
- **Faster:** Minimal overhead.
- *Use case: Live streaming, gaming, VoIP, DNS.*

UDP Segment Header (8 Bytes)

UDP is encapsulated by IP, so we primarily care about ports and data integrity.

Bit (0 - 15)	Bit (16 - 31)
Source Port (16)	Destination Port (16)
Length (16)	Checksum (16)
Application Layer Data (Size varies)	

Header Fields:

- **Src/Dst Port:** Identifies the sending and receiving processes.
- **Length:** Total length of header and data.
- **Checksum:** Basic error detection.

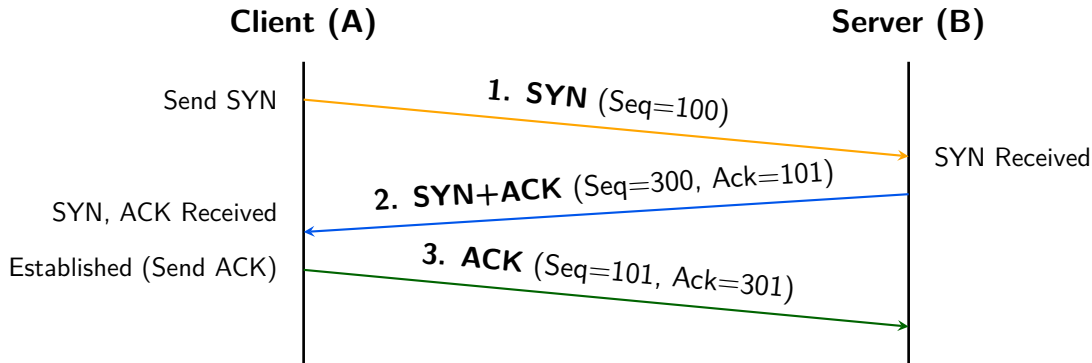
TCP Segment Header (20 Bytes)

TCP requires more fields to ensure connection state and reliability.

Bit (0 - 15)		Bit (16 - 31)	
Source Port (16)		Destination Port (16)	
Sequence Number (32)			
Acknowledgement Number (32)			
H. Length (4) / Rsvd (6)	Control Bits (6)	Window Size (16)	
Checksum (16)		Urgent Pointer (16)	
Options (0 or 32 if any)			
Application Layer Data (Size varies)			

TCP 3-Way Handshake

TCP requires establishing a connection before transferring any application data.



Reliability in Transport Layer

Reliability ensures data arrives correctly and in order at the destination. It comes with a cost in time and channel usage.

Mechanisms Utilized

- **Sequencing:** Numbers ensure packets are reassembled correctly.
- **Acknowledgments (ACK):** Receiver confirms successful receipt.
- **Retransmission:** Sender resends packet if ACK is not received before a Timeout.

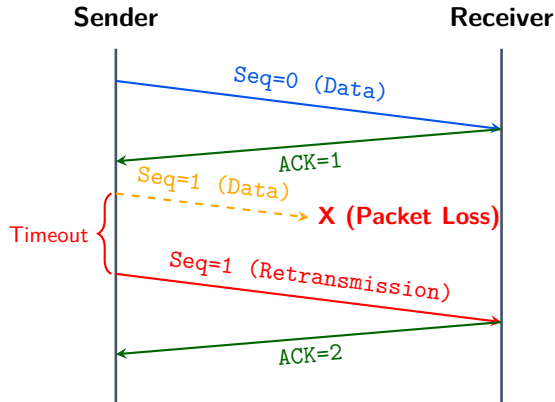
Data Transfer Phase

- Client dictates the data flows (e.g., HTTP GET Request).
- Server confirms receipt by responding with an **ACK of the Next Sequence Number**.

TCP Reliability in Action: Sequencing & ACKs

Ensuring Delivery

- **Tracking:** Sender marks each segment with a Seq number.
- **Confirming:** Receiver replies with an ACK indicating the *next* byte it expects.
- **Recovery:** If an ACK isn't received within the **Timeout** window, TCP automatically retransmits.



Example: Multiplayer Games (Minecraft & Fortnite)

Modern gaming networks use a hybrid approach to optimize experience.

When to use TCP

Used for critical game logic that *must* arrive and be processed in order.

- Player authentication & login.
- Inventory management (picking up an item).
- Chat messages.

When to use UDP

Used for real-time and high-frequency updates where speed beats perfection.

- Avatar location and movement.
- Enemy positions.
- Physics events (explosions, ballistics).

Example: Videoconferencing (Zoom)

Like gaming, communication platforms carefully partition their traffic.

- **TCP:** Used for signaling user authentication, session setup, and call negotiation (e.g., sharing encryption keys).
- **UDP:** Used for real-time streaming of encrypted audio and video. Dropping a single frame is preferable to delaying the entire video stream waiting for a retransmission.

Fallback Mechanisms

If UDP is blocked by a restrictive corporate firewall, systems like Zoom can **tunnel over TCP** or even HTTPS. The connection will succeed, but stream quality typically suffers (increased latency and jitter).

Exercise: Exploring TCP vs. UDP and Application Protocols